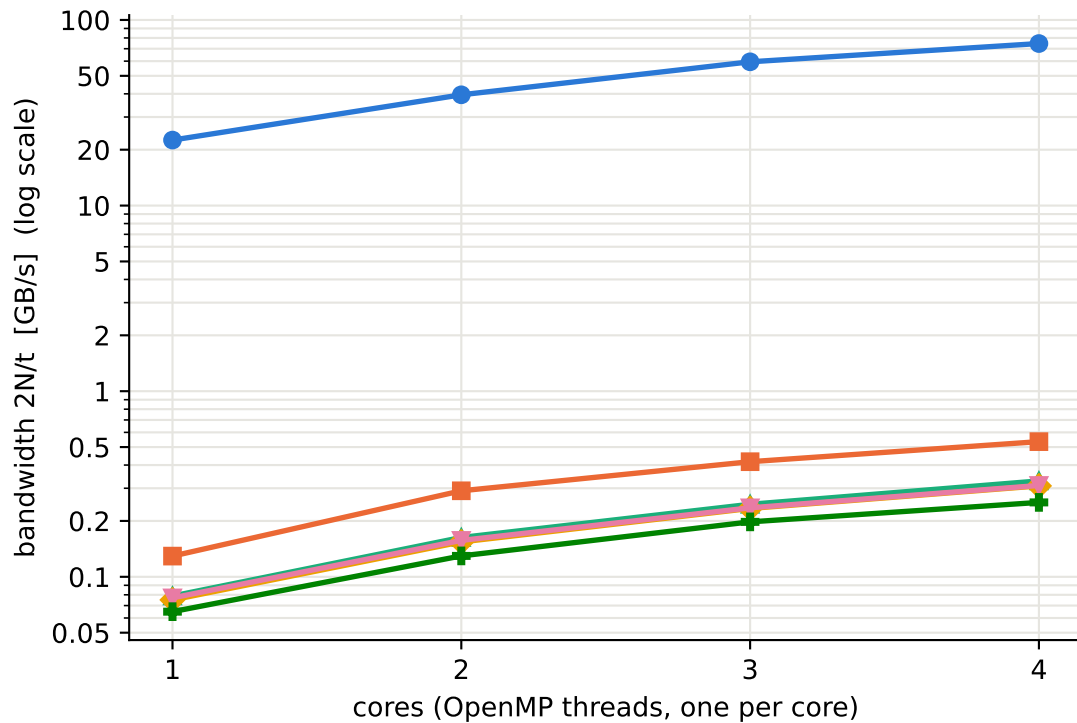
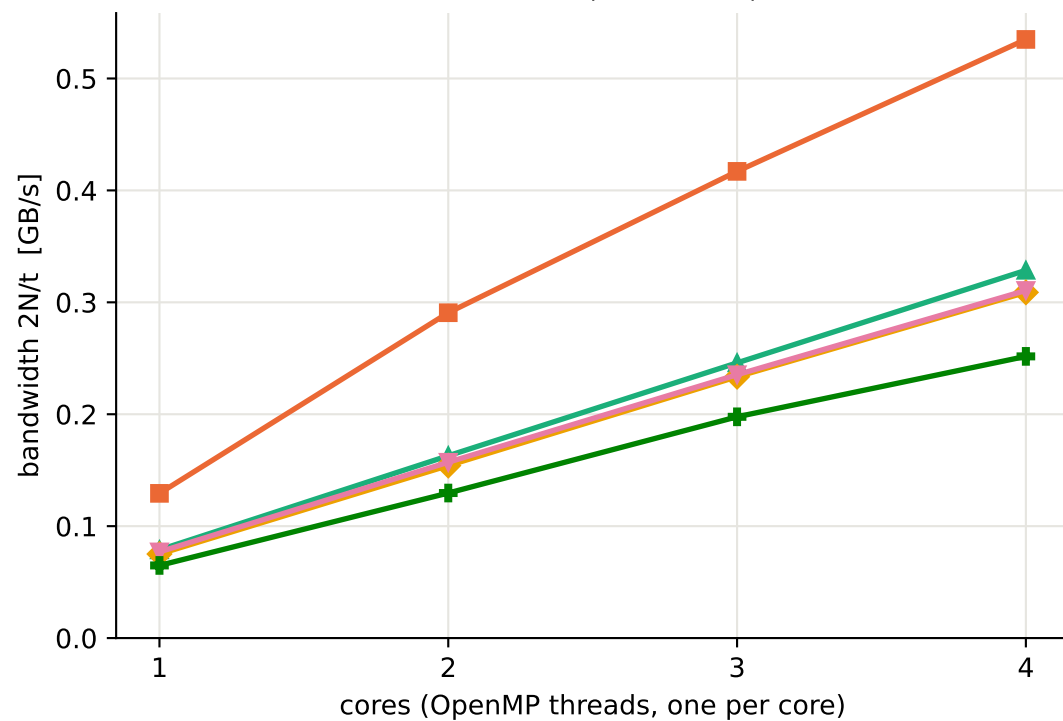


iterative: bandwidth vs cores, $N = 2^{32}$ bytes
Intel(R) Xeon(R) Processor @ 2.10GHz, 4 cores

all ranks



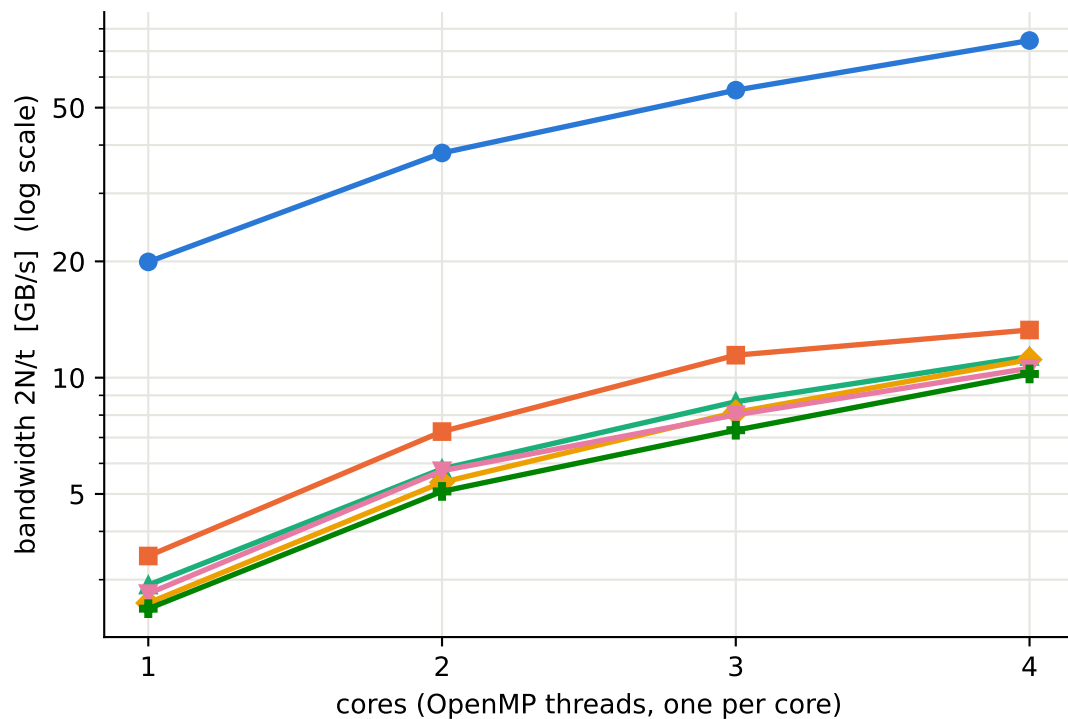
ranks ≥ 2 (linear scale)



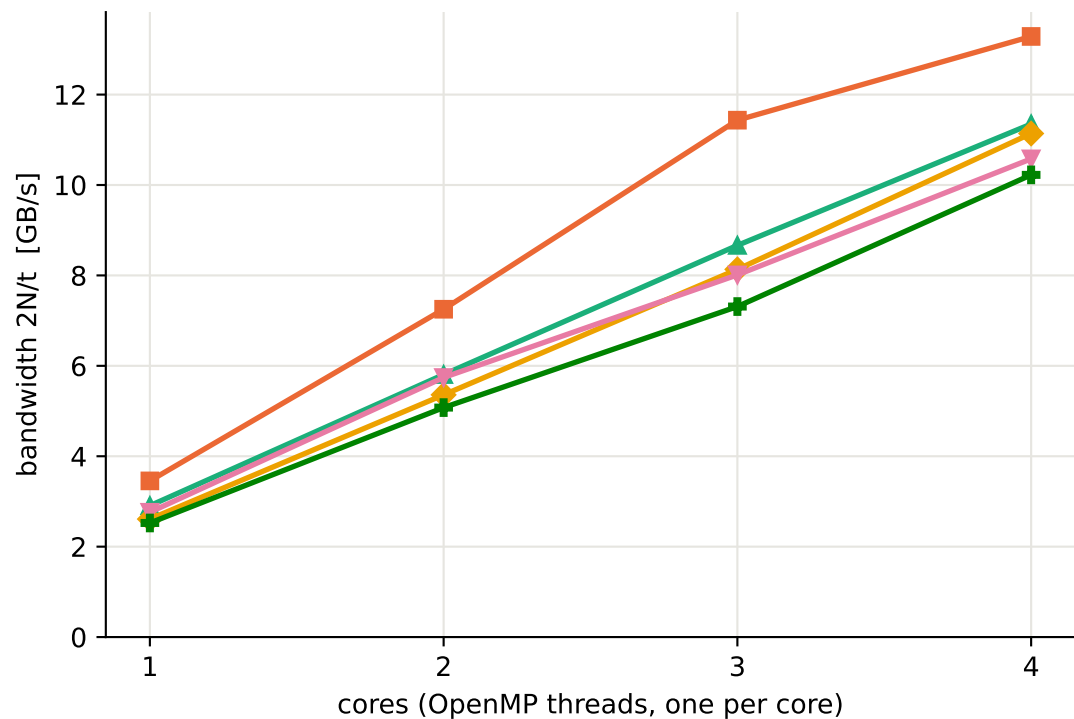
rank 1 rank 2 rank 4 rank 8 rank 16 rank 32

blocked (Part 3): bandwidth vs cores, $N = 2^{32}$ bytes
Intel(R) Xeon(R) Processor @ 2.10GHz, 4 cores

all ranks

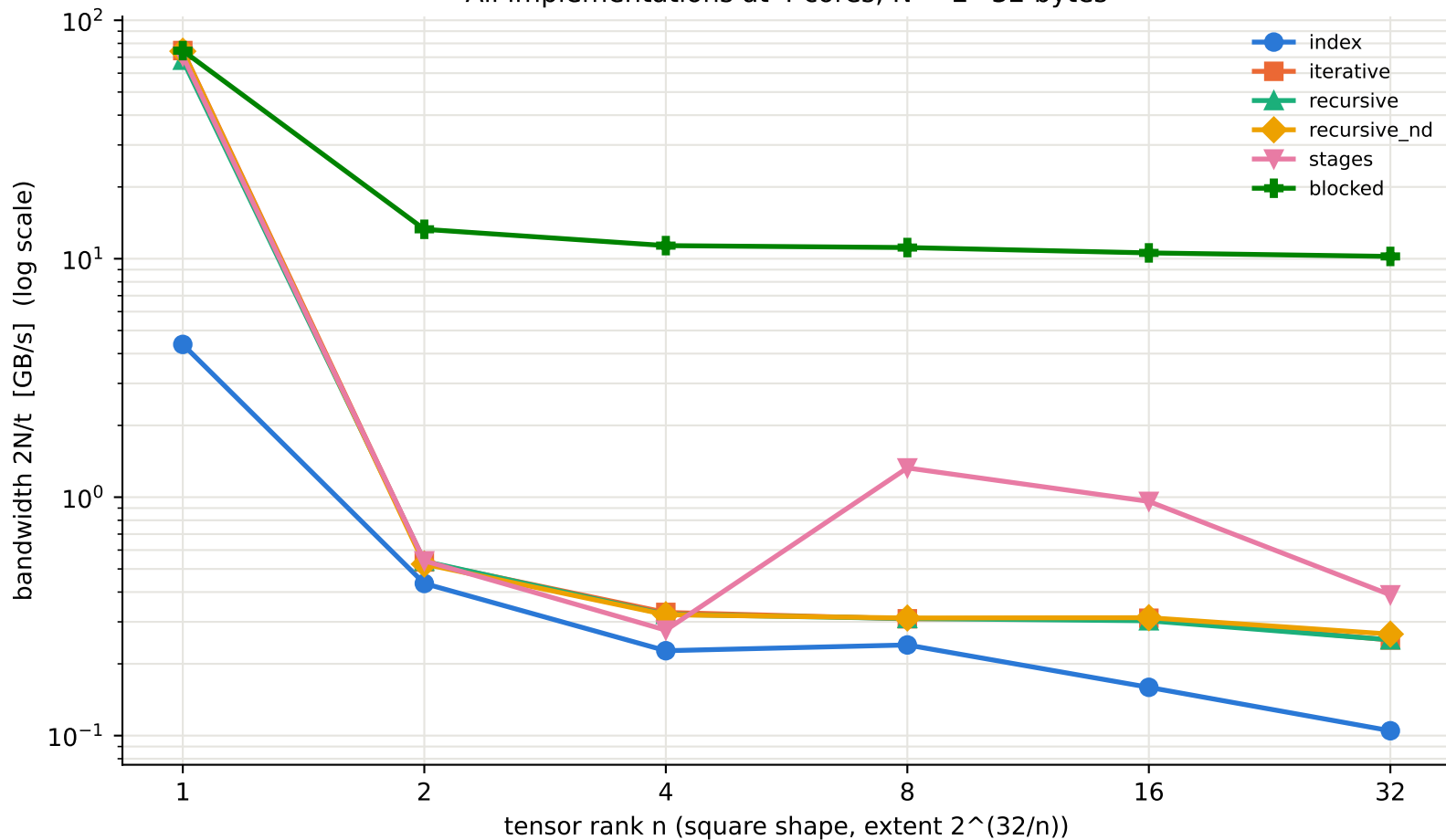


ranks ≥ 2 (linear scale)

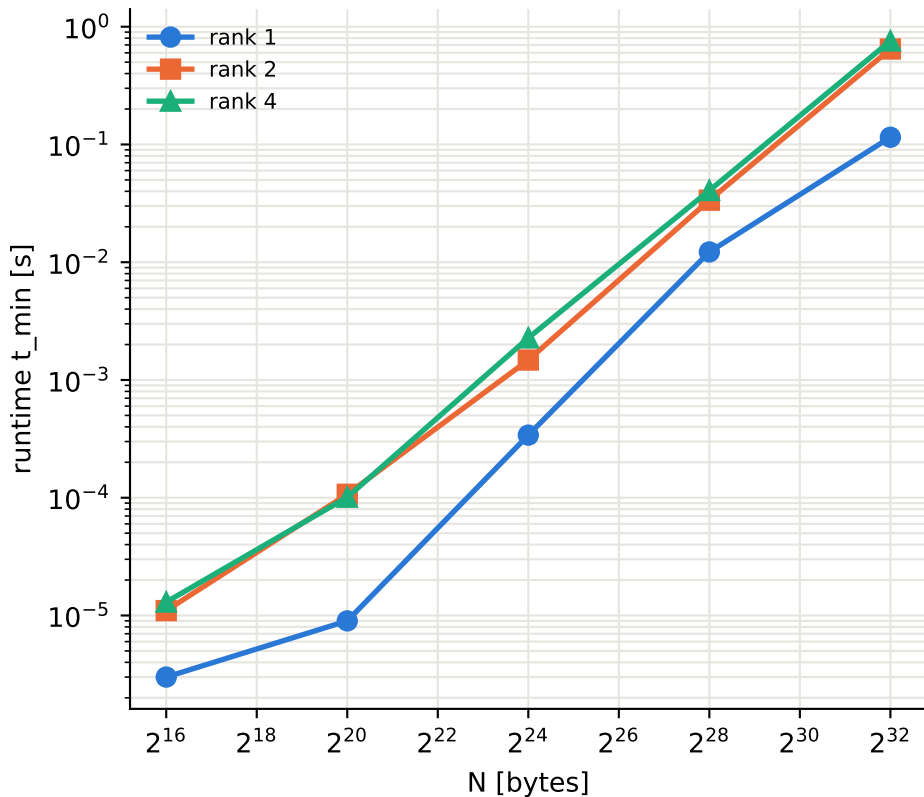


rank 1 rank 2 rank 4 rank 8 rank 16 rank 32

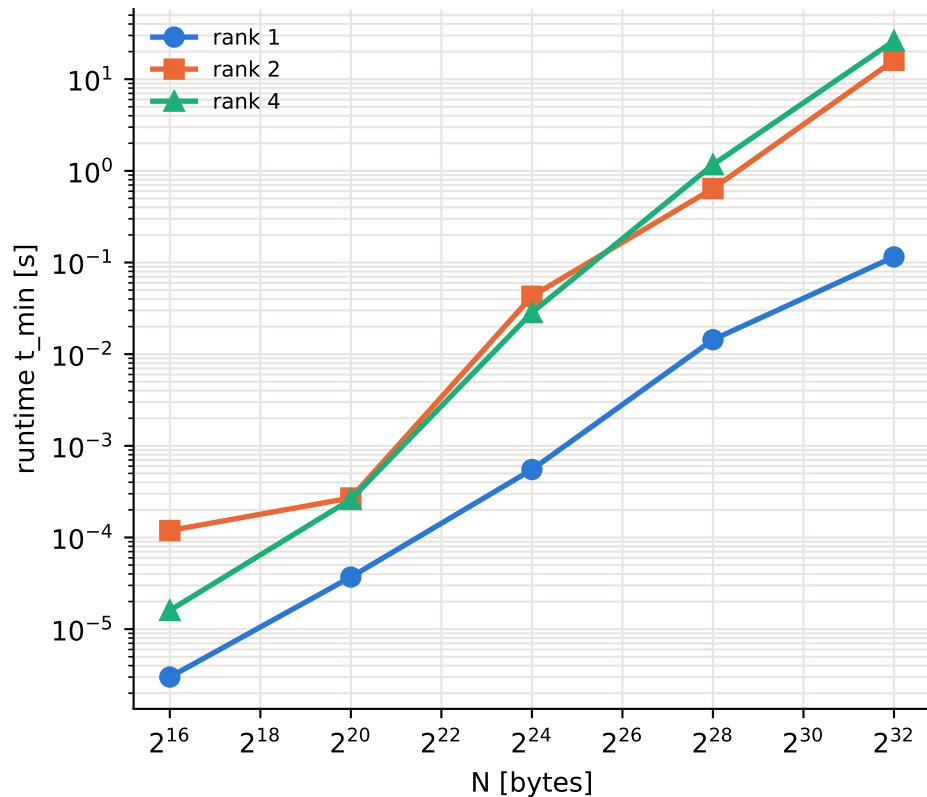
All implementations at 4 cores, $N = 2^{32}$ bytes



blocked: runtime vs problem size



iterative: runtime vs problem size



Part 3: blocking step by step (4 cores)
roof: rank-1 plain copy = 75 GB/s; unblocked iterative = 0.25-0.5 GB/s

